

Computer Networks, Spring 2026

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Lab 5: DNS Deep-Dive

In this lab, you will explore the Domain Name System (DNS) using command-line diagnostic tools and packet analysis. *This lab is to be done individually.*

1 Basic DNS Queries

Use `nslookup` and `dig` to perform basic name resolution.

1. Use `nslookup` to find the IP address of `ahduni.edu.in`.
2. Use `dig ahduni.edu.in` and identify the following sections in the output:
 - HEADER (specifically the FLAGS - is the response authoritative?)
 - QUESTION SECTION
 - ANSWER SECTION
 - AUTHORITY SECTION
3. What is the TTL (Time To Live) of the record you just received?

2 DNS Record Types

DNS supports multiple record types for different purposes.

1. Use `dig` to find the Mail Exchange (MX) records for `google.com`.
2. Use `dig` to find the Name Server (NS) records for `ahduni.edu.in`.
3. Use `dig` to find the IPv6 address (AAAA record) for `facebook.com`.

3 The Resolution Process: Tracing DNS

The `+trace` switch in `dig` allows you to see the iterative resolution process.

1. Run `dig ahduni.edu.in +trace`.
2. Identify the Root Servers contacted initially.
3. Identify the TLD (`.in`) servers contacted next.
4. Identify the authoritative servers for `ahduni.edu.in`.
5. How many iterations did it take to reach the final answer?

4 Reverse DNS Lookup

Find the hostname associated with a given IP address.

1. Perform a reverse lookup for `8.8.8.8` using `dig -x 8.8.8.8`. What is the result?

5 Evaluation

- ☐ Can interpret the flags in a DNS header. TA: _____
- ☐ Understands the difference between recursive and iterative resolution. TA: _____
- ☐ Successfully traced a resolution to the root servers. TA: _____